BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Algorithms: Examination Techniques

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How do IDE error diagnostics help?

2. Which type suits a count of pupils?

A) Integer
B) Real
C) Boolean
D) Character

3. "A character data type can hold an entire arbitrary-length sentence as one character." Correct this. Explain why.

4. A bill is quantity x unit price + fee. "Calculations always run left to right." Correct this. How does operator precedence affect the result?

Today's objectives

Explain the core ideas of algorithms: examination techniques using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Read algorithm details

Distinguish a stated loop boundary, index and test from what the algorithm is intended to do.

Trace faithfully

Record each update and branch, including false conditions and early stops.

Identify prerequisites

Explain when searching and sorting methods are valid.

Link method to marks

An exam explanation should specify precise comparisons, movement or changes to stored data.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Read algorithm details	Distinguish a stated loop boundary, index and test from what the algorithm is intended to do.
Trace faithfully	Record each update and branch, including false conditions and early stops.
Identify prerequisites	Explain when searching and sorting methods are valid.
Link method to marks	An exam explanation should specify precise comparisons, movement or changes to stored data.

I Do - One worked example

Trace linear search for 9 in [4,6,9,2] with zero-based indexes.

1. Compare index 0: 4 · 9.
2. Compare index 1: 6 · 9.
3. Compare index 2: 9 = 9; stop.
4. Found at index 2 after three comparisons.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Trace binary search for 14 in sorted [2,6,10,14,18,22,26].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the index to print the Program Counter abbreviation.

```
array registers = ["MAR","MDR","PC","ACC"]
print(registers[...])
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. A linear search seeks 7 in [3,5,7,9]. Give number of checks. [2 marks]

Q3. State a requirement of binary search. [1 mark]

Q4. Explain what one pass of bubble sort guarantees. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Linear search can work on an unsorted list. ☐ T ☐ F
2. Distinguish a stated loop boundary, index and test from what the algorithm is intended to do. ☐ T ☐ F
3. Record each update and branch, including false conditions and early stops. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 They show errors and often their location, helping the programmer find the cause. ☐

Q2 A - Integer ☐

Q3 A character represents one character; a string represents a sequence of characters. ☐

Q4 Multiplication comes before addition, giving 10 in the original example. Operator precedence affects the result. Use brackets to make the intended order clear. ☐

Code First - Model

```
array registers = ["MAR","MDR","PC","ACC"]
print(registers[2])
```

Output: PC

We Do - Check the model

1. Index 3 found with one comparison. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 3. ☐

Q3 Sorted input. ☐

Q4 After a full ascending pass, the largest unsorted value has reached the right end of that section. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: